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RESEARCH ARTICLE

Stackelberg Game-Theoretic Safe MARL With Bilevel Control for Autonomous Driving

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ABSTRACT Ensuring safety in interactive autonomous driving remains a core challenge for reinforcement learning (RL), since agents must act under uncertainty and rare but critical events (e.g., collisions) while respecting traffic rules such as yielding and right-of-way. To address these challenges, we propose **DiStaK**, a distributional Stackelberg RL framework that models driving interactions as a bilevel leader–follower game. A practical discrete instantiation, **DiStaK-C51**, augments the safety layer with a C51-based cost head to estimate the full distribution of safety costs and constructs chance-constrained admissible action sets via cumulative distribution (CDF) thresholding. To improve efficiency, DiStaK-C51 replaces exhaustive joint-action Stackelberg enumeration with a retriever–refiner Top-K'/Top-K selection rule: a lightweight retriever produces a small candidate list, chance-constraint screening filters unsafe actions, and a final Top-K shortlist supports critic-based refinement. The follower selects a risk-aware best response using $Q_2 - \lambda_2 \bar{C}_2$ with an adaptive dual update on λ_2, and leader actions can be screened based on the induced interaction outcome, with a relaxation fallback to avoid deadlock when estimated safe sets are empty. We evaluate DiStaK-C51 on standard *two-vehicle* merge and roundabout benchmarks, where it achieves substantially improved safety metrics while maintaining strong task performance and stable learning dynamics. We also provide theoretical analysis showing that the (fallback-augmented) screened safe Stackelberg Bellman operator is a contraction and that Top-K shortlisting and distributional projection yield an explicit ϵ-neighborhood bound. Finally, we outline a practical extension to multi-vehicle traffic via rule-based role assignment and a horizontal two-level Stackelberg expansion, while comprehensive multi-vehicle evaluation is left for future work.

INDEX TERMS Autonomous driving, bi-level optimization, reinforcement learning, Stackelberg game.

I. INTRODUCTION

Autonomous driving systems (also known as autopilots) must make rapid and reliable decisions in dynamic, interactive traffic environments [1], [2], [3]. These decisions (e.g., whether to yield, merge, or overtake) require not only physical control but also social awareness and compliance with implicit driving conventions [4], [5], [6]. As traffic density and heterogeneity increase, decision-making and control become core challenges for ensuring both safety and efficiency. Despite substantial progress in perception and planning, designing a decision framework that adapts to uncertain interactions while providing practical safety assurances remains an open research problem [7], [8].

With advances in artificial intelligence and vehicle engineering, three mainstream paradigms have evolved: rule-based, optimization-based, and learning-based (particularly reinforcement learning, RL) approaches [9], [10]. Rule-based methods are simple, interpretable, and stable in structured scenarios such as car-following or lane keeping [11], [12]. Yet, their finite rule sets struggle to capture the diversity and stochasticity of real-world driving behaviors, leading to brittle responses under unstructured or rare situations. Optimization-based methods formulate decision-making as a constrained

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optimization problem, elegantly encoding objectives such as comfort, efficiency, and safety [13], [14]. However, their reliance on accurate dynamics and explicit cost modeling limits adaptability. In highly interactive settings, latency and model mismatch can degrade online performance, especially when facing human drivers' unpredictable strategies.

Human driving often follows a hierarchical logic of right-of-way and yielding, making game-theoretic modeling particularly well-suited. The Stackelberg leader–follower framework provides a principled formulation for asymmetric interactions: one vehicle (leader) commits to an action, and the other (follower) responds conditionally [15]. This decomposition captures decision order and compliance behavior observed in real traffic. When augmented with probabilistic safety budgets, unsafe or non-compliant actions can be filtered early in the decision process—an appealing property for interpretable and regulation-aware autonomous behavior.

Building on this foundation, reinforcement learning (RL) offers a natural way to operationalize such hierarchical interactions through experience-driven policy optimization. Unlike rule- or optimization-based methods that rely on pre-specified cost functions or analytic solvers, RL learns strategic behavior from environment feedback [16]. In a Stackelberg context, both leader and follower policies can be trained to reason about one another, allowing adaptive driving behaviors to emerge from interaction rather than manual rule design. Meanwhile, multi-agent reinforcement learning (MARL) provides a general perspective for modeling many-vehicle traffic networks and has motivated a broad literature on coordination and scalability [16], [17].

Despite these advantages, practical adoption of Stackelberg-based RL in *discrete* action spaces remains limited due to computational and modeling bottlenecks. Joint leader–follower evaluation exhibits combinatorial expansion, which can induce high variance targets and learning instability. Moreover, discretization may lead to sensitivity under near-tied action values, resulting in policy jitter or switching behavior. In addition, uncertainty calibration for safety-critical outcomes (e.g., rare collisions) is non-trivial when the safety signal is sparse and long-tailed. Therefore, this paper focuses on discrete Stackelberg driving interactions and aims to develop a feasible and scalable decision mechanism with explicit safety screening.

To overcome these limitations, we formulate autonomous-driving decision-making as a **sublinear distributional Stackelberg reinforcement learning (DiStaK-RL)** problem, integrating hierarchical interaction and uncertainty within a unified framework. The approach couples a **distributional cost head** with a **bilevel safety constraint** to construct adaptive safe action sets from estimated cost distributions. To balance efficiency and safety, we replace exhaustive joint evaluation with an **adaptive Stackelberg selection rule (Top-K best response)**, shifting response computation from multiplicative dependence on the full joint action space to near-linear dependence on the leader's action size and a small candidate set. Under a unified paradigm for **discrete** decision spaces, **DiStaK-C51** applies Top-K conditional responses with *selective* best-response updates under distributional safety constraints, improving tractability and supporting near real-time Stackelberg reasoning in interactive traffic.

The formulation and empirical evaluation in this manuscript focus on *two-vehicle* leader–follower interactions (merge and roundabout), where the leader/follower roles are determined by scenario-specific right-of-way rules (e.g., mainline priority in merging and circulating-traffic priority in roundabouts). To address denser traffic with $N > 2$ vehicles, we provide in later sections a formal *horizontal* multi-vehicle extension: multiple leaders act first (possibly simultaneously) and multiple followers respond conditionally on the leader action profile, while preserving DiStaK-C51's "distributional safety filter Top-K response" mechanism. Multi-vehicle empirical evaluation is left for future work.

A. CONTRIBUTIONS

This paper makes the following contributions. (1) We propose **DiStaK**, a *distributional* Stackelberg RL framework for interactive driving, where safety is enforced by *chance-constrained* admissible sets constructed from predicted cost distributions (CDF thresholding). (2) We introduce a **sublinear** discrete Stackelberg selection rule (**retriever–refiner Top-$K'K$**) that couples candidate retrieval, distributional safety screening, and Top-K refinement, replacing exhaustive Stackelberg enumeration. (3) We provide an operator-theoretic analysis of the (fallback-augmented) screened safe Stackelberg Bellman operator and derive explicit error-tube bounds induced by Top-K shortlisting and distributional projection. (4) We empirically validate the framework on merge and roundabout benchmarks and outline a practical multi-vehicle extension via right-of-way role assignment and horizontal Stackelberg expansion.

II. RELATED WORK

A. CONVENTIONAL CONTROL AND PLANNING

Early autonomous driving research primarily focused on rule-based and model-based control frameworks. Paden et al. [9] provide a comprehensive survey of motion planning and control techniques, highlighting deterministic planning algorithms and model predictive control (MPC) for urban vehicles. Scarinci and Heydecker [11] examined motorway merging, proposing control concepts that rely on predefined vehicle behaviors to ensure safe interactions. Chen et al. [12] and Zhou et al. [14] extended these ideas using bionic and event-triggered MPC methods for ramp merging and path tracking, demonstrating effectiveness in simulator validations. These approaches offer high interpretability and stability under well-modeled scenarios, but their rigidity limits adaptability to dynamic or uncertain environments, particularly in mixed-traffic conditions.

Simulation studies further revealed the impact of human driver behavior on traditional control systems. Aramrattana et al. [18] investigated platooning gaps, showing that conventional drivers react unpredictably to merging vehicles. Such findings highlight the necessity for adaptive strategies that can accommodate heterogeneous traffic agents.

B. REINFORCEMENT LEARNING FOR AUTOPILOTS

Reinforcement Learning (RL) has become a dominant paradigm for learning adaptive driving policies in complex environments. Foundational works, such as Mnih et al. [19] and Silver et al. [20], demonstrated that deep RL can learn high-dimensional control strategies, inspiring numerous autonomous driving applications [13], [16], [17], [21]. Ran et al. [22] and Lin et al. [23] applied the DQN and DDQN algorithms to the lane-changing strategy and adaptive steering control strategy, demonstrating that these methods have good adaptability in automatic control scenarios involving discrete tasks. Wang et al. [24] and Likmeta et al. [25] further explored hybrid approaches combining rule-based constraints with RL, improving policy transparency and safety compliance.

Deep RL methods have been applied in urban and highway scenarios to handle mixed traffic. Irshayyid et al. [10], [26] and Wang et al. [27] demonstrated multi-agent RL for platoon merging and mixed traffic flow, emphasizing coordination among autonomous and human-driven vehicles. These studies highlight RL's strength in learning adaptive policies where traditional planning fails, yet they also reveal challenges: high sample complexity, unstable convergence, and difficulty ensuring safety guarantees.

C. GAME AND STACKELBERG MODELS

Game-theoretic frameworks provide a structured way to model interactive decision-making. Stackelberg games, introduced in classical economic theory [15], have been increasingly adopted in autonomous driving to model leader-follower interactions [28], [29]. By explicitly considering hierarchical dependencies, these methods capture right-of-way and yielding behaviors in dense traffic, offering interpretable interaction models.

Ji et al. [28] applied Stackelberg games to lane-merging strategies, demonstrating improved efficiency in dense traffic. Niu et al. [29] extended this approach for continual policy improvement in scenario-based closed-loop driving. While these methods provide theoretical guarantees of rational behavior, they are computationally intensive and often rely on perfect rationality assumptions, which may not hold in realistic, uncertain traffic scenarios.

D. SAFE REINFORCEMENT LEARNING IN MULTI-AGENT SYSTEMS

The combination of multi-agent reinforcement learning (MARL) and safe learning has emerged as a promising direction. Dinneweth et al. [16] and Zhang et al. [17] survey MARL applications in autonomous vehicles, highlighting challenges in coordination, scalability, and partial observability. Variable-noise MADDPG [30] and hierarchical TD3 [31] integrate game-theoretic reasoning and attention mechanisms, allowing agents to account for interactions while optimizing policies in continuous action spaces.

Safe MARL methods address safety-critical constraints by incorporating bilevel optimization or Lagrangian duality. Zheng and Gu [32] proposed a bilevel safe MARL framework, ensuring constraint satisfaction in autonomous driving tasks. Araia [33] applied chance-constrained RL to balance risk and performance under uncertainty. Ding et al. [34] developed provably efficient generalized Lagrangian policy optimization for multi-agent settings, enabling theoretical safety guarantees during training. These approaches advance the field by explicitly considering safety and compliance but may still face computational challenges when scaling to dense traffic networks. Meanwhile, recent work has also explored integrating explicit rule-based risk assessment and action correction into RL for lane change decision-making, improving safety by screening or adjusting risky actions during learning and deployment [35].

E. DISTRIBUTIONAL REINFORCEMENT LEARNING AND UNCERTAINTY MODELING

Distributional RL (DistRL) captures the return distribution [36], [37], which help improving robustness and risk-sensitive decision-making [38], [39], [40]. In autonomous driving, DistRL can quantify uncertainty in traffic interactions, supporting safer and more robust policy selection [2], [41]. Combining DistRL with Stackelberg hierarchical modeling enables probabilistic anticipation of follower behaviors, integrating interaction-aware planning with risk-sensitive learning [31].

F. DISCUSSION

Traditional control methods for autonomous driving, including model predictive control, proportional-integral-derivative strategies, and bionic decision-making approaches [9], [11], [12], have been extensively employed to ensure interpretability and engineering transparency. These methods, while reliable in structured scenarios, lack the adaptability required to handle complex and highly dynamic traffic interactions. To overcome this limitation, reinforcement learning (RL) and multi-agent RL (MARL) techniques have been increasingly investigated [16], [17], [26]. RL-based approaches enable autonomous agents to learn optimal behaviors through trial-and-error interactions without the need for explicit labeling [19], [20], [21], [24], [26], [27], [42]. However, conventional RL methods often fail to ensure compliance with traffic rules, interpretability, and safe interaction in dense multi-agent environments [13]. The introduction of safe RL and distributional methods [31], [32], [33] partially addresses these challenges by explicitly incorporating risk awareness

and constraints, albeit at the cost of increased computational complexity.

Learning-based methods provide significant advantages in representing high-dimensional sensory inputs and adapting to varying traffic conditions. Supervised or imitation learning approaches [18], [43] can be effective when large-scale, high-quality datasets are available, yet they require substantial labeling efforts and often struggle to generalize to rare or out-of-distribution events. In contrast, RL methods allow autonomous agents to explore lane changes, merging, and intersection scenarios autonomously, facilitating adaptive decision-making [20], [21]. Nevertheless, flat RL policies without hierarchical structuring or explicit safety constraints are prone to suboptimal behaviors in interactive driving scenarios, where achieving high task return, low variance, and risk control simultaneously remains challenging [13]. Recent highway-focused surveys further emphasize that, beyond reporting learning curves, rigorous safety validation in lane change and merging tasks typically requires comprehensive safety-related metrics and evaluation protocols (e.g., collision rate and violation-related statistics) [13].

Game-theoretic frameworks, particularly Stackelberg leader–follower models, have been employed to encode structured interactions and traffic right-of-way rules [15], [28], [29]. These methods enable interpretable decision-making by defining leader and follower roles, where the follower optimizes its policy conditional on the leader's action. Such modeling naturally supports compliance with traffic rules and provides a mechanism to prune unsafe actions, thereby improving both safety and interpretability. However, scalability issues arise in discrete action spaces due to the combinatorial complexity of joint action evaluation, and standard continuous control methods such as DDPG and TD3 [44], [45] often fail to enforce feasibility or leader–follower structure in multi-agent scenarios, leading to instability and non-stationarity.

Despite these advances, several key challenges persist in interactive autonomous driving. Ensuring safety under complex interactions remains difficult, as conventional safe RL methods often rely on manually designed cost functions or constrained objectives that may not capture rare but critical collision events. Standard RL algorithms also lack mechanisms for explicit enforcement of traffic rules such as yielding or right-of-way, which can result in overly aggressive or unsafe behaviors. Moreover, scalable Stackelberg reasoning in *discrete* action spaces is hindered by the combinatorial complexity of joint leader–follower evaluation, leading to high-variance training targets and unstable learning dynamics.

To address these limitations, we propose a distributional Stackelberg reinforcement learning framework named DiStaK, which models driving interactions as a bilevel leader–follower game. A practical variant, DiStaK-C51, incorporates a C51-based cost head to estimate the full distribution of safety costs and adaptively defines safe action sets through cumulative distribution thresholds. By employing a Top-K Stackelberg response rule, exhaustive joint-action enumeration is replaced, reducing response computation from quadratic to linear with respect to the follower's action space and improving scalability. Empirical evaluations on standard *two-vehicle* merge and roundabout benchmarks demonstrate over 20% improvement in safety metrics while maintaining task performance. Theoretical analysis further establishes convergence to an ϵ-neighborhood of the optimal Bellman solution for the proposed learning dynamics. In this paper, DiStaK unifies distributional modeling, Stackelberg game theory, and scalable constrained optimization, providing a robust approach to safety, efficiency, and scalability challenges in leader–follower driving interactions. The framework of the method is shown in **Fig.** 1. We additionally provide a formal extension to multi-vehicle traffic in later sections, while multi-vehicle empirical evaluation is left for future work.

III. PRELIMINARIES AND PROBLEM FORMULATION

A. SCENARIO

In a real roundabout scenario, the green vehicle (B) acts as the Leader: it has entered and is circulating within the roundabout. The blue vehicle (A) is the follower and must yield to the leader at the entry line, proceeding only when a safe gap appears to complete its maneuver, as shown in **Fig. 2a**. In this merging scenario, the yellow vehicle (B) is the Leader travelling on the main highway. The white vehicle (A) is the Follower and must yield to the Leader, entering the main lane only when a sufficient gap is available to merge safely, as shown in **Fig. 2b**.

(a) Roundabout (b) Merging

FIGURE 1. Framework of method.

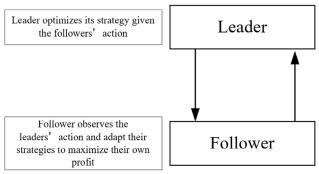


FIGURE 2. Roundabout and merging scene.

B. STACKELBERG MODEL

The Stackelberg model, originally proposed by von Stackelberg, is a cornerstone of game theory that captures sequential decision making: the leader commits first, and the follower best responds after observing the leader's move [15], the general process as shown in **Fig.** 3.

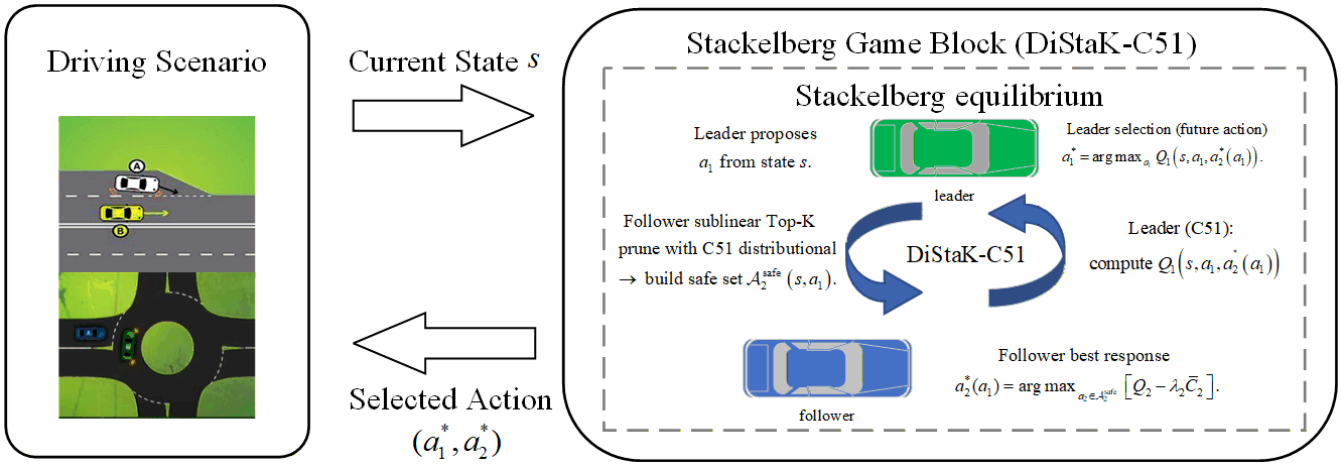


FIGURE 3. General process of Stackelberg Game.

Therefore, we can consider a two-agent Markov setting $\mathcal{G} = (\mathcal{S}, \mathcal{A}_1, \mathcal{A}_2, P, r_1, r_2, \gamma)$. Given stationary policies (π_1, π_2), the discounted return of agent $i \in \{1, 2\}$ is

$$\begin{aligned} J_i(\pi_1, \pi_2) &= \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t \, r_i(s_t, a_{1,t}, a_{2,t})\right], \\ a_{1,t} &\sim \pi_1(\cdot|s_t), \\ a_{2,t} &\sim \pi_2(\cdot|s_t, a_{1,t}). \end{aligned} \tag{1}$$

Where: $\mathcal{S}$ is the state space; $\mathcal{A}_1, \mathcal{A}_2$ are the action spaces of the leader (ego) and follower (opponent/safety layer); P is the transition kernel; r_i is the instantaneous reward of agent i; $\gamma \in (0, 1)$ is the discount factor; s_t is the global state at time t; $a_{1,t}, a_{2,t}$ are the actions sampled from π_1 and π_2 conditioned on s_t (and $a_{1,t}$ for the follower); and $\mathbb{E}[\cdot]$ denotes expectation over trajectories induced by (P, π_1, π_2).

A Markov Stackelberg equilibrium (MSE) is a pair $(\pi_1^\star, \pi_2^\star)$ such that the follower best responds to the leader and the leader anticipates this response:

$$\begin{aligned} \pi_2^\star(\pi_1) &\in \arg\max_{\pi_2} J_2(\pi_1, \pi_2), \\ \pi_1^\star &\in \arg\max_{\pi_1} J_1\big(\pi_1, \pi_2^\star(\pi_1)\big). \end{aligned} \tag{2}$$

Where: $\pi_2^\star(\pi_1)$ is the follower best-response mapping given the leader policy π_1; $\arg\max_{\pi_2}$ returns a policy attaining the maximal follower return; the leader then optimizes J_1 while *anticipating* that the follower will play $\pi_2^\star(\pi_1)$.

C. PROBLEM FORMULATION: CONSTRAINED STACKELBERG RL

Beyond efficiency and comfort, autonomous driving requires *hard safety*. Within the Stackelberg framework, we introduce long-term constraints in the spirit of CMDPs, modeling safety-related quantities (e.g., collision, minimum-TTC violation, lane-rule violation, excessive acceleration/jerk) as discounted cost sequences $c_{i,j}^t$. For agent i, the j-th long-term discounted cost is

$$G_{i,j}(\pi_1, \pi_2) = \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t \, c_{i,j}^t(s_t, a_{1,t}, a_{2,t})\right], \qquad j = 1, 2, \ldots, m. \tag{3}$$

Where: $c_{i,j}^t(\cdot)$ is the per-step safety-related cost signal j for agent i at time t (e.g., indicators of collision, minimum-TTC violation, rule violation, or excessive jerk); $G_{i,j}$ aggregates these costs with discount γ along trajectories generated by (π_1, π_2).

Given thresholds $d_{i,j}$, the *constrained* Stackelberg–bilevel RL problem is

$$\begin{aligned} \max_{\pi_1} \;\; & J_1\big(\pi_1, \pi_2^\star(\pi_1)\big) \\ & \text{s.t.} \;\; G_{1,j}\big(\pi_1, \pi_2^\star(\pi_1)\big) \le d_{1,j}, \;\; j = 1, \ldots, m_1, \\ \text{where} \;\; & \pi_2^\star(\pi_1) \in \arg\max_{\pi_2} J_2(\pi_1, \pi_2) \\ & \text{s.t.} \;\; G_{2,j}(\pi_1, \pi_2) \le d_{2,j}, \;\; j = 1, \ldots, m_2. \end{aligned} \tag{4}$$

Where: $d_{i,j}$ are safety thresholds (upper bounds) for the long-term costs; m_1, m_2 are the number of constraints enforced on the leader and follower, respectively; s.t. indicates feasibility requirements written at the *upper level* (leader) and the *lower level* (best response).

Furthermore, Eq. (4) is presented in a *general* constrained Stackelberg bilevel form, where constraints may be imposed at both the upper (leader) and lower (follower) levels. In our DiStaK-C51 instantiation, constraint enforcement is *asymmetric but two-sided* during discrete Stackelberg selection: the follower constraint is enforced *explicitly* via safe-set construction and constrained best response, while the leader constraint is applied as an additional *screening* step on the *induced interaction pair* $(a_1, a_2^\star(a_1))$ in the outer-loop leader selection. Because hard constraints at both levels can make the *joint feasible set* brittle in discrete action spaces (i.e., many states admit no admissible pair), we incorporate a relaxation fallback on the leader side: if no leader action survives the leader screening (given the constrained follower best response), we temporarily drop the leader screening for

that step, record a leader-side violation, and revert to the unconstrained Stackelberg leader choice. This preserves action coverage and stabilizes target construction/propagation, while still respecting the leader-side threshold whenever feasible.

Concretely, Section IV enforces follower safety via distributional chance-constraint screening (safe-set construction from the C51 cost head) and selects the follower response by maximizing a risk-aware score of the form $Q_2 - \lambda_2 \bar{C}_2$ with an adaptive dual update on λ_2; when the follower safe set is empty, a two-stage fallback is used to avoid deadlock. In addition, Section IV screens leader actions via the leader cost head evaluated on $(a_1, a_2^\star(a_1))$ at threshold d_1 (with budget α_1 if using a chance-constraint test); when the leader-screened set is empty, a relaxation fallback is used to preserve learning stability. The concrete constraint-enforcement mechanism and the complete algorithmic realization are presented in Section IV.

IV. METHODOLOGY (DISTAK-C51)

In multi-agent reinforcement learning, safety constraints often co-occur with hierarchical decision making, as in leader–follower interactions for autonomous driving. Two difficulties arise: (i) modelling reward and safety cost for joint actions, and (ii) in discrete spaces, forming next-step targets via enumeration or nested optimisation, which causes a combinatorial blow-up and unstable target propagation. Conventional CSQ (Constrained Stackelberg Q Learning) uses exhaustive Stackelberg selection with per-step complexity $O(|\mathcal{A}_1||\mathcal{A}_2|)$.

For discrete driving tasks, we adopt a Stackelberg formulation with chance constraints. Follower safety is enforced via a probability bound at threshold d_2, computed from a C51 follower cost head whose CDF at d_2 defines the follower safe set. In addition, we apply a leader-side chance-constraint screening at threshold d_1, evaluated on the induced interaction pair $\big(a_1, a_2^\star(a_1)\big)$, so that leader actions that lead to unsafe couplings are filtered out whenever possible. Because simultaneously constraining leader and follower can make the joint feasible set brittle under discrete Stackelberg selection, we employ a relaxation fallback on the leader side: if the leader-safe set becomes empty at a state, we temporarily drop the leader screening for that step (recording a violation) and revert to unconstrained Stackelberg selection to preserve action coverage and stabilize target propagation.

To scale selection, we replace joint enumeration with *follower-only Top-K pruning*: for each state and leader action a_1, a lightweight retriever–refiner builds a Top-K shortlist inside the follower safety set; the follower then chooses its best safe reply using the risk-aware score $Q_2 - \lambda_2 \bar{C}_2$ with an adaptive dual update on λ_2. The leader evaluates actions by reward Q_1 against that constrained best reply and selects $a_1^\star$ subject to the leader-side screening; if no leader action passes the leader screening, we relax the leader constraint for that step. The behaviour policy is ε-greedy with joint independent uniform sampling during exploration, while exploitation follows the Stackelberg order with the above constraint enforcement and fallbacks. Learning uses n-step *distributional* Bellman targets per head, projection back to a fixed atom grid, and a KL (cross-entropy) objective with slow target updates; the multiplier λ_2 is adapted from observed follower violations. Overall, under typical engineering assumptions (shared/cached forward passes and cheap retrieval), the selection workload drops to $O(|\mathcal{A}_1| \cdot K)$, yielding substantial savings for large action spaces while preserving safety consistency. The pseudocode of the discrete case is shown in Algorithm 1.

Early in training, the learned chance-constrained safe set may occasionally be empty due to prediction error. Our two-stage fallback ensures the selection operator remains well-defined (hence stable target propagation), at the expense of allowing constraint violations in those steps; we log these events and reduce their frequency via the adaptive dual update.

Prior discrete Stackelberg methods (e.g., exhaustive CSQ-style selection) not only incur $O(|A_1||A_2|)$ per-state computation, but also amplify learning instability: max-operators over large discrete joint spaces can be sensitive under near-tied values and sparse safety signals. DiStaK-C51 addresses this with a *coupled* design: (i) a distributional cost head enables *chance-based* screening by the predicted cost CDF, filtering actions whose violation probability exceeds the budget before refinement, and (ii) Top-$K'K$ shortlisting confines the subsequent argmax to a small, safety-screened set, reducing jitter and stabilizing target construction. Finally, the fallback-augmented selection keeps the operator well-defined when the estimated safe set is empty early in training (events are logged and mitigated via the adaptive dual update), which improves robustness of learning dynamics.

A. MODEL AND CHANCE CONSTRAINTS

The discrete interaction is cast as a discounted two-player Markov game; safety is expressed by chance constraints that limit the probability of violating cost thresholds.

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}_1, \mathcal{A}_2, P, r_1, r_2, c_1, c_2, \gamma), \qquad \gamma \in (0, 1). \quad (5)$$

where: $\mathcal{S}$ state space; $\mathcal{A}_1, \mathcal{A}_2$ finite action sets of leader and follower; P transition kernel; r_i reward of agent $i \in \{1, 2\}$; c_i safety cost; γ discount factor.

The following inequalities require the costs to be below thresholds with high probability.

$$\mathbb{P}\{c_i(s_t, a_{1t}, a_{2t}) \le d_i\} \;\ge\; 1 - \alpha_i, \quad i \in \{1, 2\}. \quad (6)$$

where: d_i cost boundary; $\alpha_i \in (0, 1)$ violation budget. Interpretation: the event cost $\le d_i$ must occur with probability at least $1 - \alpha_i$. In selection, follower safety (i=2) is enforced explicitly via safe-set construction, while leader safety (i=1) is applied as a screening test on the induced interaction pair with a relaxation fallback when the leader-safe set is empty.

B. C51 SETUP: ATOM GRID, LOGITS, AND EXPECTATIONS

A fixed categorical support (“atom grid”) is introduced for all distributional heads so that targets can be projected back

Algorithm 1 DiStaK-C51 (Compact): Distributional Stackelberg Selection With Top-K'/Top-K, Distillation, Adaptive λ_2, and Leader Screening

Require: Markov game $(S, A_1, A_2, P, r_1, r_2, c_1, c_2, \gamma)$; constraints (α_1, d_1), (α_2, d_2); budgets K', K, K_{retry}; ε-schedule; n, B, τ; $\lambda_{\max}$, $\eta_{\lambda,t}$.

Initialize critics θ, target $\bar{\theta} \leftarrow \theta$, retriever η, penalty λ_2, replay buffer D.
procedure FollowerBR(s, a_1) ▷ Top-K' retrieval + chance filter + Top-K refine
  $C \leftarrow \text{TopK}_{K'}\Big(s_\eta(o_2(s), a_1, a_2),\ a_2 \in A_2^{\text{feas}}(s)\Big)$.
  $B \leftarrow \{a_2 \in C :\ P_{\text{safe},2}(s, a_1, a_2) \geq 1 - \alpha_2\}$.
  if $B = \emptyset$ **then**
    $C \leftarrow \text{TopK}_{K_{\text{retry}}}\Big(s_\eta(o_2(s), a_1, a_2),\ a_2 \in A_2^{\text{feas}}(s)\Big)$.
    $B \leftarrow \{a_2 \in C :\ P_{\text{safe},2}(s, a_1, a_2) \geq 1 - \alpha_2\}$.
    if $B = \emptyset$ **then**
      $B \leftarrow A_2^{\text{feas}}(s)$; log follower fallback.
    end if
  end if
  $C_K \leftarrow \text{FirstK}(B)$ ▷ shortlist in retrieval order
  return $\arg\max_{a_2 \in C_K} \Big(Q_2(s, a_1, a_2) - \lambda_2\, \bar{C}_2(s, a_1, a_2)\Big)$.
end procedure
procedure Select(s, ε) ▷ joint ε-greedy + leader screening
  Build feasible sets $A_1^{\text{feas}}(s)$ and $A_2^{\text{feas}}(s)$.
  if Uniform$(0, 1) < \varepsilon$ **then**
    Sample $a_1 \sim A_1^{\text{feas}}(s)$ and $a_2 \sim A_2^{\text{feas}}(s)$; **return** (a_1, a_2).
  end if
  $A_1^{\text{cand}} \leftarrow \emptyset$.
  for all $a_1 \in A_1^{\text{feas}}(s)$ **do**
    $a_2^\star(a_1) \leftarrow \text{FollowerBR}(s, a_1)$.
    $V_1(a_1) \leftarrow Q_1\big(s, a_1, a_2^\star(a_1)\big)$.
    if $P_{\text{safe},1}\big(s, a_1, a_2^\star(a_1)\big) \geq 1 - \alpha_1$ **then**
      $A_1^{\text{cand}} \leftarrow A_1^{\text{cand}} \cup \{a_1\}$.
    end if
  end for
  if $A_1^{\text{cand}} \neq \emptyset$ **then**
    $a_1 \leftarrow \arg\max_{a_1 \in A_1^{\text{cand}}} V_1(a_1)$.
  else
    $a_1 \leftarrow \arg\max_{a_1 \in A_1^{\text{feas}}(s)} V_1(a_1)$; log leader fallback.
  end if
  return $(a_1,\ a_2^\star(a_1))$.
end procedure
for episode $= 1, 2, \ldots$ **do**
  Reset env; observe s_0.
  for $t = 0, 1, \ldots$ until terminal **do**
    $(a_t^1, a_t^2) \leftarrow \text{Select}(s_t, \varepsilon)$; step env; store transition in D.
    if learning step **then**
      Sample batch from D; compute n-step targets and $(s^{(n)}, d^{(n)})$.
      Obtain $(a^{1\star}, a^{2\star}) \leftarrow \text{Select}(s^{(n)}, 0)$ (no grad).
      Update C51 heads $H \in \{R_1, C_1, R_2, C_2\}$ by projection + KL;
    soft-update target: $\bar{\theta} \leftarrow (1 - \tau)\bar{\theta}\ \ \tau\theta$.
      Teacher $t \leftarrow Q_2(s, a_1, a_2) - \lambda_2\, \bar{C}_2(s, a_1, a_2)$; update retriever η by MSE regression.
      $\hat{\rho}_2 \leftarrow \frac{1}{B}\ \mathbf{1}\{P_{\text{safe},2} < 1 - \alpha_2\}$;
    $\lambda_2 \leftarrow \min\{\lambda_{\max}, \max[0, \lambda_2\ \ \eta_{\lambda,t}(\hat{\rho}_2 - \alpha_2)]\}$.
    end if
  end for
  Evaluate with $\varepsilon = 0$; record metrics.
end for

to the same support; see (7).

$$z_m = v_{\min} \ (m-1)\,\Delta z, \quad m = 1, \ldots, M,$$
$$\Delta z = \frac{v_{\max} - v_{\min}}{M-1}. \tag{7}$$

where: $\{z_m\}$ atoms; $v_{\min}, v_{\max}$ bounds; M number of atoms; Δz spacing.

Each head outputs logits, which are transformed into categorical probabilities by a softmax as in (8).

$$p_\theta^H(m \mid s, a_1, a_2) = \frac{\exp(\ell_\theta^H(s, a_1, a_2)_m)}{\sum_{k=1}^{M} \exp(\ell_\theta^H(s, a_1, a_2)_k)}. \tag{8}$$

where: ℓ_θ^H logits for head $H \in \{R_1, C_1, R_2, C_2\}$; p_θ^H probabilities on the grid $\{z_m\}$.

Scalar expectations are then exposed for fast ranking and pruning, as shown in (9a)-(9b).

$$Q_i(s, a_1, a_2) = \sum_{m=1}^{M} z_m \, p_\theta^{R_i}(m \mid s, a_1, a_2), \tag{9a}$$
$$\bar{C}_i(s, a_1, a_2) = \sum_{m=1}^{M} z_m \, p_\theta^{C_i}(m \mid s, a_1, a_2). \tag{9b}$$

here: Q_i expected return; $\bar{C}_i$ expected cost. Interpretation: while $Q_i, \bar{C}_i$ are used for ranking, it is the full distribution that supports probability safety below.

C. SAFETY PROBABILITY AND SAFE SET

The probability that the interaction remains within the safety bound is the cumulative distribution function of the predicted cost at d_i, as in (10).

$$P_{\text{safe}}^{C_i}(s, a_1, a_2) = \sum_{m:\, z_m \le d_i} p_\theta^{C_i}(m \mid s, a_1, a_2), \qquad i \in \{1, 2\}. \tag{10}$$

where: $P_{\text{safe}}^{C_i}$ safety probability for agent i; d_i cost threshold.

Follower actions whose violation probability is within budget form the follower safe set (11).

$$\mathcal{A}_2^{\text{safe}}(s, a_1) = \left\{ a_2 \in \mathcal{A}_2^{\text{feas}}(s) : P_{\text{safe}}^{C_2}(s, a_1, a_2) \ge 1 - \alpha_2 \right\}. \tag{11}$$

where: $\mathcal{A}_2^{\text{safe}}(s, a_1)$ follower chance-constrained admissible set; α_2 follower violation budget. Leader safety (i=1) is enforced by screening leader actions after computing the induced follower response, i.e., requiring $P_{\text{safe}}^{C_1}\big(s, a_1, a_2^\star(o_2(s), a_1)\big) \ge 1 - \alpha_1$, with a relaxation fallback when the leader-safe set is empty.

D. CENTRALIZED VALUE FUNCTIONS

The temporal evolution of reward and cost under joint actions is characterized with centralized critics, as in (12)-(13).

$$Q_i(s, a_1, a_2) = \mathbb{E}_{s' \sim P}\Big[r_i(s, a_1, a_2) \ \gamma \, \mathbb{E}_{a_1', a_2'} \, Q_i(s', a_1', a_2') \Big] \tag{12}$$
$$G_i(s, a_1, a_2) = \mathbb{E}_{s' \sim P}\Big[c_i(s, a_1, a_2) \ \gamma \, \mathbb{E}_{a_1', a_2'} \, G_i(s', a_1', a_2') \Big] \tag{13}$$

where: (a_1', a_2') next-step actions determined by the Stackelberg selection below. Interpretation: these are standard Bellman recursions under the selection-induced policy.

E. RETRIEVER–REFINER TOP-K′/K PRUNING FOR STACKELBERG SELECTION

To achieve sublinear follower search within the Stackelberg selection, we adopt a *retriever–refiner* design. For each leader action $a_1 \in \mathcal{A}_1$, a bi-encoder retriever produces a Top-K' candidate set; we then enforce chance-constraint feasibility and truncate to a Top-K shortlist for critic-based refinement. Here, o_2 denotes the follower's local observation (a subvector of the global state s). For each agent $i \in \{1, 2\}$ and state $s \in \mathcal{S}$, we define a feasible subset

$$A_i^{\text{feas}}(s) \subseteq A_i \tag{14}$$

that removes actions violating environment/kinematic/traffic-rule feasibility at state s (e.g., lane-change when the adjacent lane does not exist, invalid speed changes, etc.). When no additional feasibility filter is applied, $A_i^{\text{feas}}(s) = A_i$.

1) BI-ENCODER RETRIEVAL (TOP-K′ CANDIDATES)

We define a bi-encoder retriever via a query embedding $u_\eta(\cdot)$, an item embedding $v_\eta(\cdot)$, and an inner-product score:

$$u_\eta(o_2, a_1) = f_\eta\big([o_2, \text{onehot}(a_1)]\big) \in \mathbb{R}^d,$$
$$v_\eta(a_2) = E_\eta[a_2] \in \mathbb{R}^d,$$
$$s_\eta(o_2, a_1, a_2) = u_\eta(o_2, a_1)^\top v_\eta(a_2).$$

where: $f_\eta(\cdot)$ is the query encoder with parameters η, $E_\eta[\cdot]$ is an embedding table over follower actions, $[\cdot, \cdot]$ denotes concatenation, and onehot$(\cdot)$ denotes a one-hot encoding.

The Top-K' candidate set is obtained by retrieving the K' highest-scoring follower actions:

$$\mathcal{C}^{(K')}(o_2, a_1) = \text{TopK}_{K'} \left\{ a_2 \in \mathcal{A}_2^{\text{feas}}(s) : \ s_\eta(o_2, a_1, a_2) \right\}. \tag{15}$$

where: $\text{TopK}_{K'}\{\cdot\}$ returns the subset of K' actions attaining the largest values of the scoring function.

2) CHANCE-CONSTRAINT FILTERING AND TOP-K SHORTLISTING

Using the distributional cost head (C51), we compute the chance-constraint satisfaction probability $P_{\text{safe}}^{C_2}(o_2, a_1, a_2) = \mathbb{P}(C_2 \le d_2)$ via the predicted cost CDF at the threshold d_2. We then filter retrieved candidates by the violation budget α_2:

$$\mathcal{B}(o_2, a_1) = \left\{ a_2 \in \mathcal{C}^{(K')}(o_2, a_1) : \ P_{\text{safe}}^{C_2}(o_2, a_1, a_2) \ge 1 - \alpha_2 \right\}. \tag{16}$$

where: $\alpha_2 \in (0, 1)$ is the tolerated violation probability, and $P_{\text{safe}}^{C_2}$ is obtained by summing the predicted probability mass on C51 atoms not exceeding d_2 (i.e., evaluating the predicted CDF at d_2).

To bound refinement cost, we truncate the feasible list to a Top-K shortlist (in retrieval order):

$$\mathcal{C}_K(o_2, a_1) = \text{FirstK}\big(\mathcal{B}(o_2, a_1)\big), \qquad |\mathcal{C}_K(o_2, a_1)| \le K. \tag{17}$$

where: FirstK$(\cdot)$ retains the first K elements according to the retriever ranking induced by s_η, so that Top-K' controls candidate *recall* while Top-K bounds the refinement budget.

If $B(o_2, a_1) = \emptyset$, we retry with K_{retry}; if it is still empty, we relax to $C_K(o_2, a_1) = \mathcal{A}_2^{\text{feas}}(s)$ and record a follower violation.

3) REFINER BEST RESPONSE AND LEADER SELECTION (STACKELBERG BLOCK)

On the shortlist, the follower chooses a risk-aware best response using a scalarized objective:

$$a_2^\star(o_2, a_1) \in \arg \max_{a_2 \in \mathcal{C}_K(o_2, a_1)} \Big[Q_2(o_2, a_1, a_2) - \lambda_2 \, \bar{C}_2(o_2, a_1, a_2) \Big], \qquad \lambda_2 \geq 0. \tag{18}$$

where: $Q_2(\cdot)$ denotes the follower reward critic, $\bar{C}_2(\cdot)$ denotes the follower cost surrogate (computed as the mean of the distributional cost head), and λ_2 is a nonnegative penalty coefficient controlling the reward–cost trade-off.

Finally, the leader evaluates all $a_1 \in \mathcal{A}_1^{\text{feas}}(s)$ (no Top-K' pruning on the leader side) and selects by reward *subject to* a leader-side chance-constraint screening evaluated on the induced follower response:

$$a_1^\star(s) \in \arg \max_{a_1 \in \mathcal{A}_1^{\text{feas}}(s)} Q_1\big(s, a_1, a_2^\star(o_2(s), a_1)\big) \quad \text{s.t. } P_{\text{safe}}^{C_1}\big(s, a_1, a_2^\star(o_2(s), a_1)\big) \geq 1 - \alpha_1. \tag{19}$$

where: $Q_1(\cdot)$ denotes the leader reward critic, and $P_{\text{safe}}^{C_1}$ is obtained by summing the predicted probability mass on C51 atoms not exceeding d_1 (i.e., evaluating the predicted CDF at d_1). If the leader constraint yields an empty admissible set, we relax to $\mathcal{A}_1^{\text{feas}}(s)$ for that step and record a leader violation to avoid deadlock.

4) RETRIEVER TRAINING VIA CRITIC-BASED DISTILLATION

To align retrieval ranking with the refiner objective, we distill the retriever toward a teacher score:

$$t(o_2, a_1, a_2) = Q_2(o_2, a_1, a_2) - \lambda_2 \, \bar{C}_2(o_2, a_1, a_2), \tag{20}$$

where: $t(\cdot)$ is a scalar teacher target derived from the current critics, using the same risk-aware scalarization as in (18).

The retriever parameters η are trained by pointwise regression on replay samples:

$$\mathcal{L}_{\text{ret}}(\eta) = \mathbb{E}\Big[\big(s_\eta(o_2, a_1, a_2) - t(o_2, a_1, a_2)\big)^2 \Big]. \tag{21}$$

where: the expectation is taken over tuples (o_2, a_1, a_2) sampled from the replay buffer, and the teacher target is treated as a constant with respect to η.

The coefficient $\lambda_2 \geq 0$ controls the trade-off between reward maximization and cost aversion in the follower response and can be treated as a fixed hyperparameter or updated adaptively based on empirical violation statistics.

F. BEHAVIOR POLICY AND REPLAY

Equation (22) specifies a *joint* ε-greedy behavior policy aligned with Algorithm 1. At each state s, with probability ε we sample a joint action pair by independent uniform sampling over feasible action sets; with probability $1 - \varepsilon$ we exploit the deterministic Stackelberg selection, i.e., we compute the leader action $a_1^\star(s)$ from Eq. (19) together with the follower best response $a_2^\star(s, a_1^\star(s))$ from Eq. (18) based on the Top-K' retrieval, chance-constraint filtering, Top-K shortlisting, and leader-side screening described in Section IV-E. During evaluation, we set $\varepsilon = 0$.

$$\mathbb{P}_\beta(a_1, a_2 \mid s) = \varepsilon \, \mathcal{U}_{\mathcal{A}_1^{\text{feas}}(s)}(a_1) \mathcal{U}_{\mathcal{A}_2^{\text{feas}}(s)}(a_2) \quad (1 - \varepsilon) \, \delta_{a_1^\star(s)}(a_1) \, \delta_{a_2^\star(s, a_1^\star(s))}(a_2). \tag{22}$$

where $\delta_x(y)$ is the Kronecker delta (1 iff $y = x$), $\mathcal{U}_S$ denotes the uniform distribution on a finite set S, and $\mathcal{A}_i^{\text{feas}}(s) \subseteq \mathcal{A}_i$ denotes the feasible action subset for agent i at state s (if no feasibility filtering is used, $(\mathcal{A}_i^{\text{feas}}(s) = \mathcal{A}_i)$. This joint exploration enlarges coverage over feasible actions and mitigates selection bias, while exploitation preserves the leader–follower Stackelberg structure.

G. C51 N-STEP DISTRIBUTIONAL BELLMAN

An n-step signal is accumulated per head to form the distributional target.

$$g_H = {}_{j=0}^{n-1} \gamma^j \, u_H^{(tj)}, \qquad u_H \in \{r_1, r_2, c_1, c_2\}. \tag{23}$$

where: u_H is the per-step quantity (reward or cost) matched to head H; g_H is its n-step accumulation.

The distributional Bellman update moves each atom by g_H plus the discounted old atom and clips to the support, as in (24).

$$\tilde{z}_{H,m} = \text{clip}\left(g_H \;\; \gamma^n z_m, \;\; v_{\min}, \;\; v_{\max}\right). \tag{24}$$

where: $\tilde{z}_{H,m}$ shifted location for atom index m; $[v_{\min}, v_{\max}]$ categorical support.

H. PROJECTION OPERATOR ONTO THE FIXED SUPPORT

To compare the shifted distribution with the fixed grid, mass is projected back to the two nearest atoms by linear interpolation, as shown in (25). This operation guarantees that the resulting target distribution lives on the same categorical support and that total probability is conserved.

$$\begin{aligned} b_m &= \frac{\tilde{z}_{H,m} - v_{\min}}{\Delta z} \\ \ell_m &= \lfloor b_m \rfloor, \quad u_m = \lceil b_m \rceil \\ p'_H(\ell_m) &\leftarrow p'_H(\ell_m) \;\; p_H(m)\big(u_m - b_m\big) \\ p'_H(u_m) &\leftarrow p'_H(u_m) \;\; p_H(m)\big(b_m - \ell_m\big) \end{aligned} \tag{25}$$

where: $p_H(m)$ current probability at atom z_m for head H; $p'_H(\cdot)$ projected target distribution; ℓ_m, u_m indices of the two nearest neighbor atoms; Δz atom spacing; b_m the fractional coordinate of $\tilde{z}_{H,m}$ between neighbors. Interpretation: the mass at z_m is split proportionally to the distances to the two adjacent bins; probability mass is preserved and numerical stability is improved.

For compact notation, the projection may be summarized as the composition of the n-step distributional Bellman operator with the categorical projector, as in (26).

$$p'_H = \Pi_{\mathcal{Z}}\left(\mathcal{T}_H^{(n)} p_H\right). \tag{26}$$

where: $\mathcal{T}_H^{(n)}$ n-step distributional Bellman operator for head H; $\Pi_{\mathcal{Z}}$ projection to the grid $\mathcal{Z} = [v_{\min}, v_{\max}]$.

I. KL (CROSS-ENTROPY) OBJECTIVE AND TARGET UPDATES

The heads are trained to match the projected targets by a KL objective; target networks are updated slowly for stability.

$$\mathcal{L}(\theta) = \quad {}_{H\in\{R_1,C_1,R_2,C_2\}} w_H \,\mathrm{KL}\big(p'_H \,\|\, p_H(\cdot \mid s, a_1, a_2; \theta)\big). \tag{27}$$

where: w_H per-head weight (cost heads may be emphasized); p'_H target from (28). Interpretation: minimizing KL (equivalently cross-entropy up to constants) encourages the predicted categorical distribution to match the projected target on the fixed support.

$$\theta'_H \leftarrow \tau\,\theta_H \;\; (1-\tau)\,\theta'_H, \qquad \forall H. \tag{28}$$

where: τ Polyak coefficient; (θ_H, θ'_H) online/target parameters. Interpretation: slow exponential averaging reduces bootstrapping oscillations.

1) DUAL UPDATE FROM EMPIRICAL VIOLATIONS

The mini-batch violation rate for the follower is estimated and the multiplier is updated accordingly.

$$\hat{\rho} = \frac{1}{B} \; {}_{b=1}^{B} \; 1\left\{P_{\text{safe}}^{C_2}\left(s^{(b)}, a_1^{(b)}, a_2^{(b)}\right) < 1-\alpha_2\right\}. \tag{29}$$

where: B batch size; $P_{\text{safe}}^{C_2}$ from (10).

$$\lambda_2 \leftarrow \min\Big\{\lambda_{\max},\; \max\big\{0,\; \lambda_2 \;\; \eta_\lambda(\hat{\rho}-\alpha_2)\big\}\Big\}. \tag{30}$$

where: η_λ step size; $\lambda_{\max}$ clipping cap. Interpretation: if the observed violation rate exceeds the budget, the penalty increases to steer the leader toward safer pairs.

J. COMPUTATIONAL COMPLEXITY OF DISTAK-C51

We analyze the per-state cost of the discrete Stackelberg selection block in (14)–(19): Top-K' candidate retrieval on the follower side (15), chance-constraint filtering (16) and Top-K shortlisting (17), follower risk-aware best response (18), and the leader's exact outer loop over $\mathcal{A}_1$ (19) (no Top-K' pruning on the leader side).

1) FOLLOWER SIDE (FIXED A_1)

For a fixed leader action a_1, the follower executes: (i) Top-K' retrieval over $\mathcal{A}_2$ by the bi-encoder score (14), (ii) probability-safety filtering by the CDF test at d_2 (15), and (iii) Top-K refinement and final selection by the risk-aware score (16)–(17). The resulting per-a_1 cost is

$$\begin{aligned} T_{\text{fol}}(s, a_1) = {} & \underbrace{\mathcal{O}\big(|\mathcal{A}_2| \log K'\big)}_{\text{Top-}K'\text{ retrieval}} \\ & \underbrace{\mathcal{O}(K')}_{\text{prob.-safety filter at } d_2} \\ & \underbrace{\mathcal{O}(K)}_{\text{shortlist scan/final arg max}} \\ & \underbrace{\mathcal{O}\big((K' \;\; K)\, c_2\big)}_{\text{follower heads } (Q_2,\, \bar{C}_2)} . \end{aligned} \tag{31}$$

where: $|\mathcal{A}_2|$ is the follower action-set size; $K' \geq K$ are the pre-retrieval and shortlist sizes in (14) and (16); c_2 is the unit cost of evaluating the follower score $Q_2(s, a_1, a_2) - \lambda_2 \bar{C}_2(s, a_1, a_2)$ once (reward head plus cost head). The Top-K' retrieval term assumes heap-based selection among $|\mathcal{A}_2|$ scored actions; the probability-safety filter term evaluates the cost CDF at threshold d_2 for each of the K' candidates (cf. (10)–(11)); the $\mathcal{O}(K)$ term corresponds to scanning the Top-K shortlist to obtain the final arg max in (17).

2) LEADER SIDE (OUTER LOOP; CF. (18))

The leader evaluates all $a_1 \in \mathcal{A}_1$ and, for each a_1, calls Q_1 once at the selected $a_2^\star(s, a_1)$ from (17):

$$T_{\text{lead}}(s) = \underbrace{|\mathcal{A}_1|\, c_1}_{\text{one } Q_1 \text{ at } (a_1, a_2^\star)} \quad \underbrace{{}_{a_1\in\mathcal{A}_1} T_{\text{fol}}(s, a_1)}_{\text{delegate to follower block}} . \tag{32}$$

where: $|\mathcal{A}_1|$ is the leader action-set size and c_1 is the unit cost of one forward evaluation of $Q_1(s, a_1, a_2^\star)$.

3) TOTAL PER-STATE COST (STRICT BOUND, NO SHARING)

Combining (31) and (32) yields the conservative upper bound

$$\begin{aligned} T_{\text{total}}(s) = \mathcal{O}\Big(& |\mathcal{A}_1|\, c_1 \\ & |\mathcal{A}_1|\, [\, |\mathcal{A}_2| \log K' \;\; K' \;\; K \,] \\ & |\mathcal{A}_1|\, (K' \;\; K)\, c_2 \Big). \end{aligned} \tag{33}$$

4) ENGINEERING REGIME (SHARED TRUNK/STRONG RETRIEVER)

With shared feature trunks and caching across candidate evaluations, and when $K', K \ll |\mathcal{A}_2|$ so that the shortlist refinement dominates the practical workload, the headline simplifies to

$$T_{\text{total}}(s) = \tilde{\mathcal{O}}\Big(|\mathcal{A}_1|\,(K' \;\; K) \;\; |\mathcal{A}_1|\, c_1\Big) \approx \tilde{\mathcal{O}}\big(|\mathcal{A}_1|\, K\big). \tag{34}$$

where: $\tilde{\mathcal{O}}(\cdot)$ hides logarithmic factors (e.g., the $\log K'$ term from maintaining the Top-K' heap), and the approximation uses $K \leq K'$ and constant-time per-action scoring under shared trunks.

Therefore, compared with exhaustive Stackelberg selection of complexity $\mathcal{O}(|\mathcal{A}_1||\mathcal{A}_2|)$, DiStaK-C51 reduces the per-state selection cost to $\tilde{\mathcal{O}}(|\mathcal{A}_1|K)$ in the regime $K', K \leq |\mathcal{A}_2|$.

K. PRACTICAL SAFETY CONSIDERATIONS AND FAILURE MODES

Our safety mechanism is *conditional* on the accuracy of the learned cost distribution: the chance constraint is enforced whenever the *estimated* safe set is non-empty; when the estimated safe set becomes empty, the fallback step explicitly relaxes the constraint (logged as a violation) to avoid deadlock and preserve stable learning. In real driving, the key assumptions may break under distribution shift (unseen traffic patterns), partial observability and sensor noise, or mismatched opponent behavior, which can miscalibrate the learned cost CDF and thus the screened safe set. Therefore, the reported safety metrics should be interpreted as empirical performance under the simulated environment, while the theoretical analysis characterizes convergence of the screened (fallback-augmented) operator and clarifies when chance-constraint satisfaction can be expected in practice.

V. EXPERIMENT ANALYSIS AND CASE STUDY

We evaluate two `highway-env` tasks—*roundabout* and *merge*—using the discrete Stackelberg controller **DiStaK-C51**. Unless otherwise noted, all results are averaged over **3 random seeds** and reported as mean ± std. For the summary tables, we compute each metric over the **last** $N = 50$ **episodes** of each run and then aggregate across seeds.

Unified hyperparameters are: actor learning rate 10^{-4}, critic learning rate 10^{-3}, dual variables' learning rate 10^{-3}; discount γ=0.95; soft target update τ=0.005; one critic update per environment step; and a replay buffer of 5×10^5 transitions. Exploration uses a minimum noise rate of 0.02 and an ϵ floor of 0.05.

For constrained control, the follower's Lagrange multiplier is initialized at 0.5, and clipped in practice to $0 \leq \lambda_2 \leq 3$. We enforce a follower chance constraint with threshold d=5 and tolerance α=0.01. **DiStaK-C51** uses 51-atom categorical heads with reward support $[-20, 20]$ and cost support $[0, 1000]$, n=5-step distributional Bellman backups, an embedding dimension of 64, and sub-linear Stackelberg selection via Top-$K'K$ pruning with K'=64, K=16.

The shortlist size K controls the refinement budget in the follower response and therefore mediates a trade-off between scalability, task performance, and safety. From the selection rule in Section IV-E (Top-K' retrieval and Top-K shortlisting) and the complexity analysis in Section IV-J, the per-state selection workload scales near-linearly with K (up to logarithmic factors), while increasing K weakly enlarges the feasible shortlist and reduces the chance that high-value feasible actions are pruned before refinement. On the theoretical side, Appendix D formalizes the Top-K approximation as a selection error δ (Eq. (A.9)), which induces an explicit asymptotic error tube that shrinks as $\delta \to 0$ (Eq. (A.11)), e.g., when K increases. In our experiments we fix $(K', K) = (64, 16)$ across tasks to use a small constant refinement budget while retaining sufficient candidate recall through Top-K' retrieval.

Task-specific reward design and environment details are provided below.

A. FAIR COMPARISON PROTOCOL (CAPACITY AND BUDGET PARITY)

To ensure that performance differences are attributable to algorithmic design rather than implementation advantages, we match both *network capacity* and *training/tuning budgets* across all methods (Bi-AC, CSQ, CS-DDQN, NO-TOPK, NO-SAFETY, ORIGINAL, and ours/DiStaK-C51).

1) NETWORK CAPACITY PARITY

All methods share the same state features and normalization pipeline. The policy/value (or Q) networks adopt a unified backbone: a two-layer MLP with widths 256–256 and ReLU activations. For methods that require safety-related estimation (e.g., cost/constraint prediction), we attach an auxiliary *cost/constraint head* with matched hidden width (i.e., comparable head capacity to the value head) to each baseline when applicable, so that the overall parameter scale remains comparable across algorithms and results are not confounded by representational capacity differences.

2) TRAINING AND TUNING BUDGET PARITY

We keep the interaction and optimization budgets identical across methods. For the *roundabout* scenario, each run uses 10,000 environment steps; for *merge*, 20,000 environment steps. We use mini-batch size 64, and an update-to-data ratio of 1 : 1 (one gradient update per environment step). All results are reported over 3 random seeds under the same evaluation protocol (periodic evaluation at a fixed interval; final performance summarized over the last 50 episodes using mean ± standard deviation across seeds).

B. MERGE

As illustrated in **Fig.** 4, the environment consists of two main lanes and one merging ramp: the leader travels along the main road toward the merge point, while the follower starts from the ramp and must complete the merge and cross the finish line without colliding with the leader or the barrier at the end of the ramp. Random noise is added to the initial positions and speeds of both vehicles.

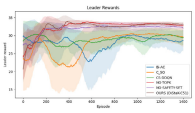
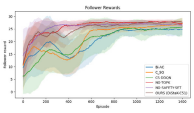
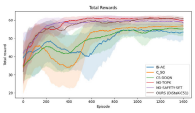
The task is a positive-sum, mixed cooperative–competitive game: on the one hand, cooperation is required to ensure a safe merge; on the other, the agents compete to reach the finish line faster for higher returns. In merge scenarios, each vehicle receives a 2 speed bonus if it maintains the same speed as the preceding vehicle. Collisions are penalized with a −5 reward. The finish bonus follows a first-come-first-served rule: the first vehicle to reach the finish line receives a 10 bonus, while subsequent arrivals receive 5.

Table 1 plots the leader, follower, and total reward curves over training episodes, averaged over 3 seeds (mean ± std). Among the baselines, **Bi-AC** converges the slowest and exhibits larger oscillations, while **CS-DDQN** improves faster but shows a pronounced mid-training dip and higher variance.

FIGURE 4. The environment of the merge.

CSQ is comparatively more stable yet plateaus at a lower total return than our method. **DiStaK-C51 (ours)** reaches a stable plateau quickly and achieves the best final *total* return with the tightest band. The exhaustive variant **Ablation-no-topK** is close in return but is computationally more expensive due to full enumeration (no Top-K pruning). Removing safety screening (**Ablation-no-safety-set**) can yield competitive rewards but comes with degraded safety outcomes (see Table. 1 and Table 2).

TABLE 1. Reward curves of merge environment across 3 seeds.

Leader	Follower	Total
		

1) SAFETY/COLLISION OUTCOMES

Fig. 5 shows the evolution of episode outcomes (leader-first arrival, follower-first arrival, and collision) across training, averaged over 3 seeds. **DiStaK-C51** rapidly drives the collision portion to near zero and maintains a near-100% arrival outcome, whereas baselines exhibit longer collision transients and larger fluctuations. The no-safety-set ablation retains a noticeably higher collision portion.

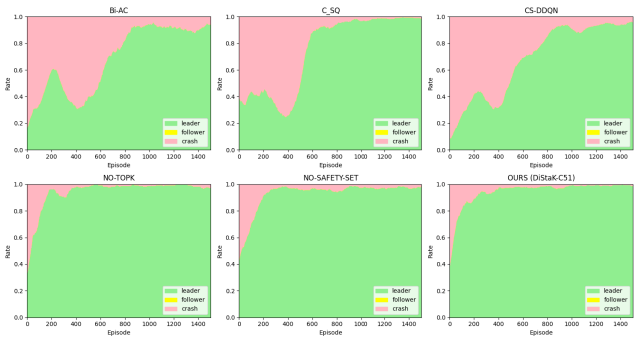


FIGURE 5. Safe vs. crash curves of different algorithms across 3 seeds.

2) FINAL PERFORMANCE

Table 2 summarizes the last $N = 50$ episodes across 3 seeds (mean $\pm$ std). **DiStaK-C51** achieves the best leader reward and total reward, with the highest arrival rate and the lowest collision rate and standard deviation. CSQ attains the highest follower reward, consistent with a more aggressive yielding/competition pattern. We also report the *empty (fallback) rate*, i.e., the frequency that the screened candidate set becomes empty and triggers fallback; this metric is only meaningful when safety screening is enabled, hence it is N.A. for the no-safety-set ablation.

Across 3 seeds, **DiStaK-C51** learns the merge task fastest and stabilizes with the highest final total return, while simultaneously achieving the best safety profile. As summarized in Table 2 over the last N=50 episodes, our method attains the best leader reward and total reward, with the highest arrival rate and *zero* collision rate. The exhaustive **Ablation-no-topK** is competitive in return but incurs higher computational cost and a noticeably higher fallback (empty-set) rate, whereas removing safety screening (**Ablation-no-safety-set**) weakens safety despite comparable rewards. Overall, DiStaK-C51 delivers the most reliable merge behavior by jointly improving efficiency, stability, and safety.

C. ROUNDABOUT

This scenario describes a circular intersection composed of two lanes (**Fig.** 6), with four entrances and exits located in the north, south, east, and west directions. In this configuration, the leader agent vehicle enters from the west entrance, while the follower agent vehicle enters from the south entrance. Given their current speeds and lane choices, the two vehicles face a potential collision risk, requiring the system to generate actions through distributed observation encoding, distributed distributional heads, and sublinear-retrieval-based Stackelberg best responses.

In the Roundabout task, the reward structure is designed to promote smooth and efficient passage through the intersection. Specifically, `collision_reward = -5` penalises any collision events. Maintaining a speed between 10–15 m/s yields a reward of 2, and an additional 5 is awarded for safely exiting the roundabout. To discourage unnecessary changing, `lane_change_reward = -0.5` penalises lane changes.

This reward design encourages cooperative behaviour and balanced control by aligning the agents' objectives with smooth navigation and safety within the roundabout.

Both vehicles follow their predefined target routes toward the northern exit; however, they must perform lane changes and longitudinal control to ensure efficient navigation within the roundabout and reduce collision probability.

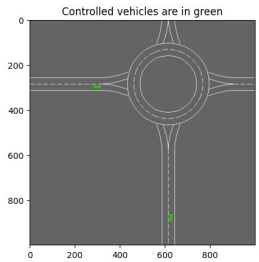


FIGURE 6. The environment of the roundabout.

In this task, the leader establishes positional advantage through first-move strategies, while the follower responds

TABLE 2. Merge scenario results over the last $N = 50$ episodes across 3 seeds (mean ± std).

Method	Leader reward	Follower reward	Total reward	Arrived rate	Crash rate	Empty rate (fallback rate)
BI-AC	29.34±4.38	26.04±3.76	55.38±6.82	0.953±0.211	0.0467±0.211	N.A.
CSQ	27.75±5.75	**28.97±4.09**	56.72±2.74	0.993±0.081	0.0067±0.081	N.A.
CS-DDQN	29.66±2.86	25.04±5.29	54.70±6.42	0.916±0.278	0.0840±0.278	N.A.
Ablation-no-topK	32.36±2.05	26.88±2.21	59.24±3.75	0.988±0.109	0.0120±0.109	0.0847±0.0978
Ablation-no-safety-set	32.16±2.29	26.54±2.78	58.70±4.40	0.973±0.161	0.0270±0.161	N.A.
Ours (DISTAK-C51)	**32.90±1.00**	27.46±1.44	**60.36±2.10**	**0.9967±0.0405**	**0.0033±0.0405**	**0.0397±0.0610**

TABLE 3. Roundabout scenario results over the last $N = 50$ episodes across 3 seeds (mean ± std).

Method	Leader reward	Follower reward	Total reward	Arrived rate	Crash rate	Empty rate (fallback rate)
BI-AC	33.20±9.45	29.00±6.94	62.20±15.59	0.900±0.300	0.100±0.140	N.A.
CSQ	35.75±6.16	31.06±4.83	66.81±9.60	0.973±0.161	0.0267±0.161	N.A.
CS-DDQN	36.62±6.27	**31.80±4.35**	68.42±10.29	0.967±0.180	0.0330±0.180	N.A.
Ablation-no-topK	38.39±8.61	26.34±5.93	64.73±12.80	0.972±0.165	0.0280±0.165	0.0973±0.133
Ablation-no-safety-set	37.93±7.62	30.24±4.92	68.17±11.33	0.960±0.196	0.0400±0.196	N.A.
Ours (DISTAK-C51)	**47.34±7.27**	24.66±4.80	**72.00±8.09**	**1.00±0.000**	**0.000±0.000**	**0.0636±0.0969**

under the constraints of safety filtering and the adaptive λ_2 mechanism. The two vehicles must cooperate to ensure safe passage through the roundabout, while simultaneously competing to secure exit priority and achieve higher rewards.

Table 4 plots the leader, follower, and total reward curves over training episodes, averaged over 3 seeds (mean ± std). Overall, **DiStaK-C51 (ours)** achieves the highest *total* return and the strongest leader reward with a relatively stable plateau. Among the baselines, **Bi-AC** exhibits slower improvement and larger oscillations, while **CSQ** and **CS-DDQN** learn faster but converge to a lower overall return than our method. The exhaustive variant **Ablation-no-topK** remains competitive but settles at a lower total return and is computationally more expensive due to removing Top-K pruning. Removing safety screening (**Ablation-no-safety-set**) can retain comparable reward levels in parts of training, but it sacrifices safety robustness (see Fig. 7 and Table 3).

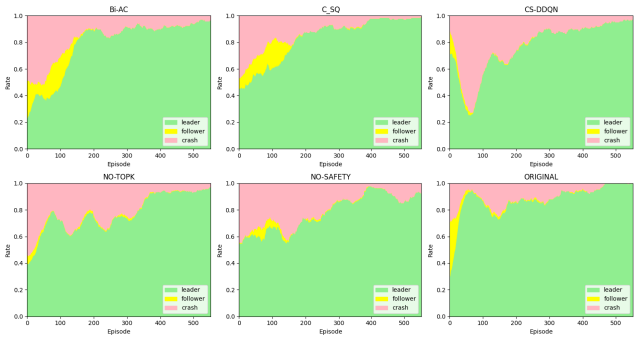
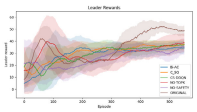
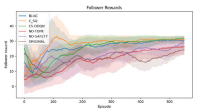
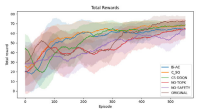


FIGURE 7. Safe vs. crash curves of different algorithms across 3 seeds.

1) SAFETY AND EPISODE OUTCOMES

Fig. 7 shows the evolution of episode outcomes (leader/follower arrival vs. collision) across training. **DiStaK-C51** reduces the collision portion rapidly and maintains a high arrival portion earlier and more consistently than the baselines. In contrast, baselines exhibit longer collision transients and larger fluctuations, and the **Ablation-no-safety-set** variant retains a noticeably larger collision portion throughout training.

TABLE 4. Reward curves of roundabout environment across 3 seeds.

Leader	Follower	Total
		

2) FINAL PERFORMANCE

Table 3 summarizes results over the last N=50 episodes across 3 seeds (mean ± std). **DiStaK-C51** achieves the best leader reward and total reward, together with the highest arrival rate and *zero* collision rate. While **CS-DDQN** attains the highest follower reward, our method delivers the highest overall return under the Stackelberg (right-of-way) setting, where the follower may yield more conservatively to guarantee safety. We also report the empty (fallback) rate for methods with screening enabled; **DiStaK-C51** exhibits a lower fallback frequency than **Ablation-no-topK**, indicating more effective candidate recall and refinement under Top-$K'K$ pruning.

Overall, these results indicate that **DiStaK-C51** provides the most reliable roundabout behavior by jointly improving efficiency, stability, and safety: it attains the highest total return with consistent convergence, eliminates collisions in steady state, and maintains a low fallback rate under screened Stackelberg responses.

VI. MULTI-VEHICLE EXTENSION VIA HORIZONTAL STACKELBERG EXPANSION

This paper primarily formulates and evaluates DiStaK-C51 under *two-vehicle* leader–follower interactions (merge and roundabout). In dense traffic, a vehicle may interact with multiple surrounding vehicles simultaneously. To bridge this gap while preserving the original DiStaK-C51 mechanism, we present a practical *horizontal* multi-vehicle extension (two-level, multi-leader/multi-follower Stackelberg), consistent with bilevel MARL formulations for autonomous driving [32]. The extension retains the key DiStaK-C51 components: (i) deterministic leader/follower assignment based on right-of-way, (ii) *distributional* chance-constraint screening via C51 cost heads, and (iii) retriever–refiner Top-K'/Top-K best-response selection to avoid exhaustive enumeration.

A. ROLE ASSIGNMENT UNDER TRAFFIC RIGHT-OF-WAY

We partition the set of interacting vehicles into a leader group $\mathcal{L}$ and a follower group $\mathcal{F}$ using scenario-specific right-of-way rules. In a merge, vehicles on the mainline are assigned as leaders while ramp vehicles are assigned as followers. In a roundabout, vehicles with priority (e.g., circulating vehicles) are assigned as leaders and entering/yielding vehicles are assigned as followers, following the traffic-rule convention of the environment. When right-of-way is ambiguous (e.g., symmetric encounters or near-simultaneous arrivals), we adopt a deterministic tie-breaking rule to ensure a well-defined hierarchy (e.g., fixed priority by relative bearing, lane index, or vehicle ID). Roles may be re-evaluated online as the interaction evolves.

B. HORIZONTAL TWO-LEVEL MULTI-VEHICLE STACKELBERG STRUCTURE

Let s denote the joint state and let A_i be the discrete action set of vehicle i. We consider a two-level Stackelberg interaction where all leaders act first (possibly simultaneously), and all followers respond conditionally. Denote $|\mathcal{L}| = n_\ell$ and $|\mathcal{F}| = n_f$. Leaders output an action profile

$$a^\ell := (a_1^\ell, \ldots, a_{n_\ell}^\ell), \tag{35}$$

where $a_i^\ell = \mu_i^\ell(o_i^\ell)$ and o_i^ℓ is the leader observation. Followers condition on (s, a^ℓ) and output

$$a^f := (a_1^f, \ldots, a_{n_f}^f), \qquad a_j^f = \mu_j^f(o_j^f, a^\ell), \tag{36}$$

where o_j^f is the follower observation. Agents within the same level can be treated as simultaneous decision-makers; the cross-level dependence remains Stackelberg since followers explicitly condition on the leader action profile a^ℓ.

1) FEASIBLE ACTION SUBSETS

For each vehicle i and state s, we optionally restrict to a feasible subset $A_i^{\text{feas}}(s) \subseteq A_i$ that removes invalid/unsafe-by-construction actions under environment/kinematic rules (if no feasibility filtering is used, $A_i^{\text{feas}}(s) = A_i$).

C. DISTRIBUTIONAL SAFETY FILTERING AND TOP-K'/TOP-K CONDITIONAL RESPONSES

We extend the DiStaK-C51 safety screening and Top-K'/Top-K response mechanism by conditioning follower feasibility on (s, a^ℓ). For each follower $j \in \mathcal{F}$, DiStaK-C51 maintains distributional heads that estimate the return and safety-cost distributions under (s, a^ℓ, a_j^f). Let $C_j(s, a^\ell, a_j^f)$ denote the follower-j random safety cost.[1] Using the predicted cost distribution, define the follower-j chance-constrained safe set

$$A_j^{\text{safe}}(s, a^\ell) := \left\{ a_j^f \in A_j^{\text{feas}}(s) \;:\; \mathbb{P}\Big(C_j(s, a^\ell, a_j^f) \le d_j\Big) \ge 1 - \alpha_j \right\}, \tag{37}$$

where d_j is the cost threshold and $\alpha_j \in (0, 1)$ is the violation budget.

1) RETRIEVER–REFINER TOP-K'/TOP-K PRUNING

To avoid exhaustive evaluation over A_j, we follow the same retriever–refiner design as DiStaK-C51. A lightweight retriever produces a Top-K' candidate list (in retrieval order)

$$C_j^{(K')}(s, a^\ell) = \text{TopK}_{K'} \left\{ a_j^f \in A_j^{\text{feas}}(s) : \; s_{\eta_j}(o_j^f, a^\ell, a_j^f) \right\}, \tag{38}$$

where s_{η_j} is a follower-j retrieval score (bi-encoder or shared retriever). We then apply chance-constraint filtering on the reduced list:

$$\begin{aligned} &B_j(s, a^\ell) \\ &:= \left\{ a_j^f \in C_j^{(K')}(s, a^\ell) \;:\; \mathbb{P}\Big(C_j(s, a^\ell, a_j^f) \le d_j\Big) \ge 1 - \alpha_j \right\}. \end{aligned} \tag{39}$$

Finally, we truncate to a Top-K shortlist in retrieval order:

$$C_{j,K}(s, a^\ell) := \text{FirstK}\big(B_j(s, a^\ell)\big), \qquad |C_{j,K}(s, a^\ell)| \le K. \tag{40}$$

2) RISK-AWARE CONDITIONAL BEST RESPONSE (DISTAK-CONSISTENT)

The follower then computes a conditional best response by maximizing the same risk-aware score as DiStaK-C51:

$$a_j^{f\star}(s, a^\ell) \in \arg \max_{a_j^f \in C_{j,K}(s, a^\ell)} \Big[Q_j^f(s, a^\ell, a_j^f) - \lambda_j \overline{C}_j^f(s, a^\ell, a_j^f) \Big], \tag{41}$$

where Q_j^f is the follower-j reward critic, $\overline{C}_j^f$ is the follower-j mean-cost surrogate, and $\lambda_j \ge 0$ is a (possibly adaptive) dual penalty updated from empirical violations, analogous to the two-vehicle case.

[1] In a horizontal implementation, C_j is evaluated on the induced interaction context (leaders' actions and the current/induced follower responses). This preserves the original DiStaK-C51 mechanism while avoiding full joint enumeration over followers.

3) FALLBACKS (AVOID DEADLOCK)

When the chance-filtered list is empty, we use the same two-stage fallback as DiStaK-C51: retry with a larger K_{retry}; if still empty, relax to $A_j^{\text{feas}}(s)$ and record a violation. Equivalently, in compact form, if $B_j(s, a^\ell) = \emptyset$ after retries we set $C_{j,K}(s, a^\ell) \leftarrow A_j^{\text{feas}}(s)$ for that step.

D. LEADER SCREENING AND PRACTICAL LEADER UPDATES

In many driving setups, the leader may also use a cost threshold (as in DiStaK-C51 with leader screening). A scalable horizontal realization applies *leader-side screening* on the induced interaction outcome after followers respond: once $a^{f\star}(s, a^\ell)$ is computed, leader actions that violate a leader chance constraint (evaluated via a leader C51 cost head at threshold d_i with budget α_i) can be filtered out. If the leader-screened set becomes empty, we relax the leader screening for that step and record a violation to preserve action coverage and learning stability.

Since enumerating the full leader action profile a^ℓ scales exponentially in $|\mathcal{L}|$, a practical implementation uses a priority-ordered/sequential leader update or independent leader selection (treating other leaders' actions as part of the state or as predicted actions). This keeps the dominant selection workload linear in $_{i\in\mathcal{L}}\,|A_i|$ rather than $_{i\in\mathcal{L}}\,|A_i|$, while preserving the same "distributional safety filter Top-K'/Top-K response" structure on the follower side.

E. DISCUSSION: PRACTICALITY AND COMPLEXITY

The horizontal extension (35)–(41) preserves the key benefit of DiStaK-C51: follower responses are computed without enumerating the full follower joint action space. Specifically, each follower j refines at most K actions (plus retriever overhead) per leader context, yielding near-linear response complexity in K rather than $|A_j|$. Right-of-way role assignment yields an interpretable decomposition into leader and follower groups; conditioning followers on the leader action profile a^ℓ captures the dominant directional influence implied by traffic priority. Although empirical evaluation in this manuscript focuses on two-vehicle benchmarks, the above formulation provides a concrete and implementable pathway to extend DiStaK-C51 to multi-vehicle scenarios without altering its core "distributional safety filter Top-K'/Top-K Stackelberg response" design, while avoiding the exponential blow-up from full joint enumeration by using practical leader updates.

VII. CONCLUSION

This paper proposes a safety-focused reinforcement learning framework for highly interactive and strongly constrained traffic by combining Stackelberg game-theoretic modeling with a bilevel structure. The discrete controller, **DiStaK-C51**, uses a distributional cost head (C51) to construct chance-constrained admissible action sets via CDF thresholding, and replaces exhaustive joint-action Stackelberg search with an efficient retriever–refiner Top-K'/Top-K mechanism. The follower selects a risk-aware best response using the scalarized score $Q_2 - \lambda_2 \bar{C}_2$ with an adaptive dual update on λ_2. In addition, leader actions can be screened by a leader-side chance test evaluated on the induced interaction pair, with a relaxation fallback to preserve action coverage.

On standard *two-vehicle roundabout* and *merge* benchmarks, DiStaK-C51 achieves consistently stronger safety–reward trade-offs than competitive baselines, with smoother and faster learning dynamics; collisions are substantially reduced while high safe-pass rates are maintained in our experiments. On the theoretical side, the appendix establishes that the (fallback-augmented) screened safe Stackelberg Bellman operator is a γ-contraction under the sup norm, implying existence/uniqueness of the fixed point and linear convergence of value iteration. When Top-K shortlisting and distributional projection are used, the analysis yields an explicit error-tube (an ϵ-neighborhood) whose radius shrinks as K increases and the atom grid is refined. Chance-constraint satisfaction is guaranteed by construction whenever the estimated safe set is non-empty; when fallback is triggered, the constraint is explicitly relaxed (logged via a fallback/violation indicator) to avoid deadlock while preserving stable target propagation.

Limitations and future work. (i) For more complex traffic and larger fleets, we will further study scalable retrieval and approximate best-response schemes and empirically evaluate the multi-vehicle extension in Section VI under denser multi-car simulations, reporting comprehensive safety statistics (e.g., collision/violation rates, fallback frequency, and confidence intervals). (ii) We plan to incorporate environment uncertainty and distribution shift into the safety budget and improve calibration of the learned cost distributions. (iii) We will evaluate the method in more realistic simulators (e.g., CARLA). Overall, DiStaK-C51 provides a scalable and interpretable safety-first approach to leader–follower decision-making in interactive autonomous driving, together with a concrete pathway toward multi-vehicle traffic.

APPENDIX A
DISCRETE CONTROL: SAFE STACKELBERG VALUE ITERATION (WITH LEADER SCREENING)

In this appendix, we provide an operator-theoretic convergence argument for the *discrete* DiStaK-C51 selection rule via a contraction of a safe Stackelberg Bellman operator, following the operator viewpoint in [46]. The analysis focuses on the induced Bellman operator for the leader action-value and does not depend on distributional implementation details beyond the safe-set (CDF) screening mechanism.

A. PROBLEM, SAFE SETS, AND SCREENED OPERATOR

Let $(\mathcal{S}, \mathcal{A}_1, \mathcal{A}_2, P, r_1, r_2, c_1, c_2, \gamma)$ be a discounted two–player Markov game with $\gamma \in (0, 1)$. Fix follower chance-constraint parameters (α_2, d_2) and (optional) leader screening parameters (α_1, d_1).

1) FOLLOWER CHANCE-CONSTRAINED SAFE SET

For each $s \in \mathcal{S}$ and leader action $a_1 \in \mathcal{A}_1$, define the *follower safe set*

$$\mathcal{A}_2^{\text{safe}}(s, a_1) = \left\{a_2 \in \mathcal{A}_2 : \Pr\left[c_2(s, a_1, a_2) \le d_2\right] \ge 1 - \alpha_2\right\}. \quad \text{(A.1)}$$

(In DiStaK-C51, the probability in (A.1) is read from the CDF of the C51 cost head at level d_2; this equals the event measure of the categorical distribution over the fixed atom grid. The operator-theoretic framing below does not depend on distributional details.)

2) RISK-AWARE FOLLOWER BEST RESPONSE

Fix a nonnegative multiplier $\lambda_2 \ge 0$ (adapted online in DiStaK-C51). Define the follower's *risk-aware best response*

$$\mathrm{BR}_{2,\lambda_2}(s, a_1) \in \arg\max_{a_2 \in \mathcal{A}_2^{\text{safe}}(s,a_1)} \left\{ Q_2(s, a_1, a_2) - \lambda_2 \overline{C}_2(s, a_1, a_2)\right\}. \quad \text{(A.2)}$$

If $\mathcal{A}_2^{\text{safe}}(s, a_1) = \emptyset$, we adopt the fallback $\mathrm{BR}_{2,\lambda_2}(s, a_1) \in \arg\max_{a_2 \in \mathcal{A}_2} Q_2(s, a_1, a_2)$ (this enlarges the feasible set and does not break the contraction argument below; see Lemma 1).

3) LEADER SCREENED SET (WITH RELAXATION FALLBACK)

Given the induced follower response in (A.2), define a leader-side screening set

$$\mathcal{A}_1^{\text{scr}}(s) = \left\{a_1 \in \mathcal{A}_1 : \Pr\left[c_1\left(s, a_1, \mathrm{BR}_{2,\lambda_2}(s, a_1)\right) \le d_1\right] \ge 1 - \alpha_1\right\}. \quad \text{(A.3)}$$

If $\mathcal{A}_1^{\text{scr}}(s) = \emptyset$, we *relax* screening for that step by setting $\mathcal{A}_1^{\text{scr}}(s) \leftarrow \mathcal{A}_1$ and recording a leader-side violation (this matches the algorithmic fallback used in the method section to avoid deadlock and preserve action coverage).

4) SAFE STACKELBERG BELLMAN OPERATOR WITH LEADER SCREENING

Let $Q : \mathcal{S} \times \mathcal{A}_1 \to \mathbb{R}$ denote the leader action-value function over state–leader-action pairs. Define the operator $T : \mathbb{R}^{\mathcal{S}\times\mathcal{A}_1} \to \mathbb{R}^{\mathcal{S}\times\mathcal{A}_1}$ by

$$(TQ)(s, a_1) = r_1\left(s, a_1, \mathrm{BR}_{2,\lambda_2}(s, a_1)\right) \gamma\, \mathbb{E}_{s' \sim P(\cdot \mid s, a_1, \mathrm{BR}_{2,\lambda_2}(s,a_1))}\Big[\max_{a_1' \in \mathcal{A}_1^{\text{scr}}(s')} Q(s', a_1')\Big]. \quad \text{(A.4)}$$

All maxima are taken with a fixed tie-breaking rule. We use the sup-norm $\|Q\| = \sup_{s\in\mathcal{S},\, a_1\in\mathcal{A}_1} |Q(s, a_1)|$.

Lemma 1: [Well-posedness of T] For any Q, the right-hand side of (A.4) is finite and well-defined. Moreover, if $\mathcal{A}_2^{safe}(s, a_1) = \emptyset$, replacing $\mathrm{BR}_{2,\lambda_2}$ by the fallback $\arg\max_{a_2\in\mathcal{A}_2} Q_2(s, a_1, a_2)$ preserves all inequalities used below. Likewise, if $\mathcal{A}_1^{scr}(s') = \emptyset$, relaxing to $\mathcal{A}_1^{scr}(s') = \mathcal{A}_1$ ensures the max in (A.4) is taken over a nonempty set and preserves the contraction proof.

Rewards are bounded in standard driving simulators; if not, they can be clipped without affecting the sup-norm contraction proof. If $\mathcal{A}_2^{\text{safe}}(s, a_1) = \emptyset$, the fallback enlarges the feasible set (from safe to full), so any upper bounds derived using the safe-set maximization remain valid when the maximization is over $\mathcal{A}_2$. If $\mathcal{A}_1^{\text{scr}}(s') = \emptyset$, relaxing to $\mathcal{A}_1$ enlarges the maximization domain in (A.4); this can only increase the value of the max, and the Lipschitz bound used in Lemma 2 continues to hold because the max remains over a subset of (s', a_1') pairs bounded by the sup-norm.

B. CONTRACTION OF THE SCREENED SAFE STACKELBERG OPERATOR

Lemma 2: [Contraction] T defined in (A.4) is a γ–contraction on $(\mathbb{R}^{\mathcal{S}\times\mathcal{A}_1}, \|\cdot\|)$:

$$\left\|TQ - T\widetilde{Q}\right\| \le \gamma \left\|Q - \widetilde{Q}\right\| \qquad \forall Q, \widetilde{Q}.$$

Fix (s, a_1). Write $a_2^\star = \mathrm{BR}_{2,\lambda_2}(s, a_1)$; it depends on $(Q_2, \overline{C}_2, \lambda_2)$ (not on Q). The immediate reward terms cancel when we take differences of $(TQ)(s, a_1)$ vs. $(T\widetilde{Q})(s, a_1)$. Hence

$$\begin{aligned}
&\left|(TQ)(s, a_1) - (T\widetilde{Q})(s, a_1)\right| \\
&= \gamma \left|\mathbb{E}_{s'}\Big[\max_{a_1' \in \mathcal{A}_1^{\text{scr}}(s')} Q(s', a_1') - \max_{a_1' \in \mathcal{A}_1^{\text{scr}}(s')} \widetilde{Q}(s', a_1')\Big]\right| \\
&\le \gamma\, \mathbb{E}_{s'}\Big[\max_{a_1' \in \mathcal{A}_1^{\text{scr}}(s')} \left|Q(s', a_1') - \widetilde{Q}(s', a_1')\right|\Big] \\
&\le \gamma \left\|Q - \widetilde{Q}\right\|.
\end{aligned} \quad \text{(A.5)}$$

Taking supremum over (s, a_1) gives the claim.

Theorem 1: [Existence and uniqueness of the fixed point; linear convergence] T has a unique fixed point $Q^\star$, and the value iteration $Q^{k1} = TQ^k$ satisfies

$$\left\|Q^k - Q^\star\right\| \le \gamma^k \left\|Q^0 - Q^\star\right\| \qquad (k \ge 0).$$

Banach's fixed-point theorem follows immediately from Lemma 2.

C. THE FIXED POINT INDUCES A STATIONARY SCREENED CONSTRAINED STACKELBERG EQUILIBRIUM

Definition 1: [Greedy screened leader and risk-aware follower] Given $Q^\star$, define the follower's one-step rule $\pi_2^\star(\cdot|s, a_1)$ to be a Dirac mass at $\mathrm{BR}_{2,\lambda_2}(s, a_1)$. Define the leader's stationary greedy policy with screening

$$\pi_1^\star(\cdot|s) \in \arg\max_{a_1 \in \mathcal{A}_1^{\text{scr}}(s)} Q^\star(s, a_1),$$

where $\mathcal{A}_1^{\text{scr}}(s)$ uses the relaxation fallback $\mathcal{A}_1^{\text{scr}}(s) \leftarrow \mathcal{A}_1$ if empty.

Lemma 3: [Dynamic programming optimality] Let $Q^\star$ be the unique fixed point of T. Then for every s,

$$\max_{a_1 \in \mathcal{A}_1^{\text{scr}}(s)} Q^\star(s, a_1) = r_1\left(s, a_1^\star, \mathrm{BR}_{2,\lambda_2}(s, a_1^\star)\right) \gamma\, \mathbb{E}_{s'}\Big[\max_{a_1' \in \mathcal{A}_1^{\text{scr}}(s')} Q^\star(s', a_1')\Big], \quad \text{(A.6)}$$

where $a_1^\star \in \arg\max_{a_1 \in \mathcal{A}_1^{\mathrm{scr}}(s)} Q^\star(s, a_1)$ (with the same tie-breaking rule on both sides).
Insert $Q^\star = TQ^\star$ into (A.4), then take $\max_{a_1 \in \mathcal{A}_1^{\mathrm{scr}}(s)}$ on both sides, using the same tie-breaking order on both sides.

Theorem 2: [Screened constrained Stackelberg equilibrium (stationary)] The pair $(\pi_1^\star, \pi_2^\star)$ constructed above is a stationary screened constrained Stackelberg equilibrium for the chance-constrained problem: the follower best-responds with respect to $Q_2 - \lambda_2 \overline{C}_2$ inside its safe set and the leader is greedy w.r.t. $Q^\star$ within the screened set $\mathcal{A}_1^{\mathrm{scr}}(s)$ (with relaxation fallback when empty).
Lemma 3 shows $Q^\star$ certifies leader optimality against the risk-aware best reply within the screened domain; the follower's definition (A.2) enforces the chance constraint by construction (or relaxes only when the safe set is empty). Hence neither player can profitably deviate in a one-step dynamic programming sense; iterating over the horizon with the same best reply yields a stationary equilibrium under discounting and screening.

1) PRACTICAL IMPLICATION OF EMPTY SAFE SETS AND FALLBACK

The convergence argument above is often presented under the simplifying assumption that $\mathcal{A}_2^{\mathrm{safe}}(s, a_1) \neq \emptyset$ for all (s, a_1). In practice, especially early in training, the learned cost predictor may be inaccurate or overly pessimistic, and the estimated safe set can be empty at some states.

Our implementation resolves this by defining a *fallback-augmented* (relaxed) safe set

$$\widetilde{\mathcal{A}}_2^{\mathrm{safe}}(s, a_1) := \begin{cases} \mathcal{A}_2^{\mathrm{safe}}(s, a_1), & \mathcal{A}_2^{\mathrm{safe}}(s, a_1) \neq \emptyset, \\ \mathcal{A}_2, & \text{otherwise}, \end{cases} \tag{A.7}$$

and the corresponding best response

$$\widetilde{\mathrm{BR}}_{2,\lambda_2}(s, a_1) \in \arg \max_{a_2 \in \widetilde{\mathcal{A}}_2^{\mathrm{safe}}(s,a_1)} \left\{ Q_2(s, a_1, a_2) - \lambda_2 \overline{C}_2(s, a_1, a_2) \right\}. \tag{A.8}$$

Replacing $\mathrm{BR}_{2,\lambda_2}$ by $\widetilde{\mathrm{BR}}_{2,\lambda_2}$ yields a fallback-augmented Bellman operator $\widetilde{T}$ defined analogously to (A.4). Since the proof of Lemma 2 only uses the fact that the next-state maximization is over a nonempty set, $\widetilde{T}$ remains a γ–contraction under the same sup-norm. Therefore, the value iteration converges linearly to the unique fixed point of the *relaxed* operator $\widetilde{T}$ even when empty safe sets occur.

Importantly, the chance constraint is enforced *whenever* $\mathcal{A}_2^{\mathrm{safe}}(s, a_1) \neq \emptyset$. When $\mathcal{A}_2^{\mathrm{safe}}(s, a_1) = \emptyset$, the fallback step may violate the chance constraint; we record such events and adapt λ_2 based on empirical violation statistics, which in practice reduces the frequency of empty-safe-set states as training progresses. Thus, the theoretical guarantee should be interpreted as convergence to the fixed point of a fallback-augmented operator, with constraint satisfaction holding on the subset of states where the learned safe set is nonempty.

D. TOP-K APPROXIMATION AND DISTRIBUTIONAL PROJECTION: EXPLICIT ERROR BOUNDS

1) TOP-K SELECTION ERROR (FOLLOWER SHORTLISTING)

Suppose the follower's Top-K shortlisting introduces a uniform selection error $\delta \geq 0$: for every (s, a_1), the chosen $a_2^{(K)}(s, a_1)$ satisfies

$$\begin{aligned} &Q_2(s, a_1, a_2^{(K)}) - \lambda_2 \overline{C}_2(s, a_1, a_2^{(K)}) \\ &\geq \max_{a_2 \in \mathcal{A}_2^{\mathrm{safe}}(s,a_1)} \left(Q_2 - \lambda_2 \overline{C}_2 \right) - \delta. \end{aligned} \tag{A.9}$$

Let T_K denote the Bellman operator that uses $a_2^{(K)}$ instead of $\mathrm{BR}_{2,\lambda_2}$. Then for any Q,

$$\begin{aligned} &\|T_K Q - TQ\| \\ &\leq \underbrace{\sup_{s,a_1} \left| r_1\left(s, a_1, a_2^{(K)}(s, a_1)\right) - r_1\left(s, a_1, \mathrm{BR}_{2,\lambda_2}(s, a_1)\right) \right|}_{\leq \delta_r}. \end{aligned} \tag{A.10}$$

A standard perturbed-contraction argument yields the finite error tube

$$\begin{aligned} &\left\| Q^{k1} - Q^\star \right\| \leq \gamma \, \left\| Q^k - Q^\star \right\| \, \delta_r, \\ &\qquad \Rightarrow \quad \limsup_{k \to \infty} \left\| Q^k - Q^\star \right\| \leq \frac{\delta_r}{1 - \gamma}. \end{aligned} \tag{A.11}$$

If $\delta_r \to 0$ (e.g., $K \uparrow |\mathcal{A}_2|$ or improved retrieval/shortlisting), then $Q^k \to Q^\star$.

2) C51 PROJECTION ERROR IN EXPECTATION

Let Π_{C51} be the categorical projection onto the fixed atom grid $\{z_m\}_{m=1}^M$ with spacing Δz. For any target scalar random variable Y supported in $[v_{\min}, v_{\max}]$, the expectation error of projection obeys the triangle-weighting bound

$$\left| \mathbb{E}[\Pi_{C51} \mathsf{Law}(Y)] - \mathbb{E}[Y] \right| \leq \Delta z.$$

Hence the per-update expected target error is at most Δz; combining with (A.11) yields an asymptotic tube of radius $\Delta z(1 - \gamma)$ that vanishes as the atom grid is refined.

APPENDIX B
WHAT EACH CONVERGENCE RESULT MEANS FOR THE ALGORITHMS

For DiStaK-C51: under the standard bounded-reward assumption and with the follower safe sets $\mathcal{A}_2^{\mathrm{safe}}(s, a_1)$ defined in (A.1), the screened safe Stackelberg Bellman operator T in (A.4) is a γ–contraction. Leader screening is incorporated via $\mathcal{A}_1^{\mathrm{scr}}(s)$ in (A.3), with a relaxation fallback when the screened set is empty. Hence:

- *Exact selection (no Top-K approximation):* $Q^k \to Q^\star$ and $(\pi_1^\star, \pi_2^\star)$ is a stationary screened constrained Stackelberg equilibrium.
- *Top-K and C51 projection:* the limit error obeys the explicit tube bound (A.11); refining K and the atom grid (smaller Δz) shrinks the tube toward 0.

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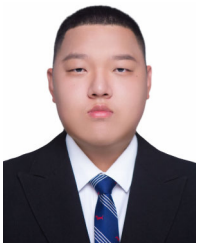

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